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Where Should I Focus My Return Reduction Efforts? Empirical Guidance for Retailers

Guangzhi Shang†
Department of Analytics, Information Systems, and Supply Chain, College of Business, Florida State University, Tallahassee, FL, 32306, e-mail: gshang@business.fsu.edu

Mark E. Ferguson
Department of Management Science, Moore School of Business, University of South Carolina, Columbia, SC, 29208, e-mail: mark.ferguson@moore.sc.edu

Michael R. Galbreth
Department of Business Analytics & Statistics, Haslam College of Business, University of Tennessee, e-mail: galbreth@moore.sc.edu

ABSTRACT

Owing to the generous refund policy ubiquitous among today's retailers, returns are prevalent and expensive to manage. Implementing buyer assistance programs and adjusting return time windows are two common methods that retailers use to actively influence return rates. A buyer assistance program can improve consumer understanding and reduce returns, but to be effective it must be focused on the products that are most likely to be returned. Managing return rates by adjusting return time windows, on the other hand, requires an understanding of the time dimension of returns, a topic for which there is scant guidance in the literature. Using a large dataset of consumer electronics purchases and returns for a major U.S. retailer, our study contributes to the research and practice of retail operations by: (i) providing new insights into the drivers of product returns and (ii) analyzing the time length of consumers' product trial, which can inform return time window decisions. We show that both product maturity (where a product stands in its life-cycle) and variety (the number of competing alternatives) are negatively associated with the probability of a return. By separating pure returns and exchanges, we further demonstrate that while more variety considerably reduces the former, it only marginally increases the latter. In addition, we find that consumers with more past returns not only have a higher probability of future returns but also tend to make late returns, with both factors increasing a retailer's cost to serve them. Finally, our analysis of the time to return shows that retailers may benefit from customizing their return time windows across different product categories, as well as enlarging the window during holidays.[Submitted: January 31, 2018. Revised: August 10, 2018. Accepted: September 12, 2018.]

Subject Areas: Consumer Returns, Econometric Analysis, and Retail Operations.

†Corresponding author.

INTRODUCTION

One of the biggest challenges for U.S. retailers is managing the considerable volume of consumer returns. In 2017, the National Retail Federation estimates that the total value of consumer returns was around $351 billion in the United States, an increase of 89% from the 2007 estimate (National Retail Federation, 2008, 2018). Overall, retail return rates between 10% and 20% have been observed in recent years (Tuttle, 2011; National Retail Federation, 2018). Generous return policies, nearly ubiquitous among big name retailers, are a primary contributor to this state of consumer returns. The key motivation behind a generous return policy comes from the demand-side benefits. Because consumers are often uncertain about the value of the products they intend to purchase, the option to return allows them to mitigate the negative consequences if they decide, after some trial period, that they do not like a product. Anderson, Hansen, and Simester (2009) estimate that, for catalog apparel items, a full refund policy increases a customer's utility of an item by 35% of the purchase price. Similarly, Rao, Lee, Connelly, and Iyengar (2018) demonstrate that consumers are willing to pay more when the return time window is longer. In addition, as shown in Griffis, Rao, Goldsby, and Niranjan (2012) and Petersen and Kumar (2009), consumers who make more returns in the past tend to purchase more frequently and buy higher value items in the future. Thus, a generous return policy, such as a full money back guarantee, is believed to entail price and demand benefits for the retailer.

These demand-side marketing benefits have come at a steep cost on the operational side, however. The handling and disposition costs of returns account for 5–6% of a typical (Original Equipment Manufacturer) OEM's revenue and 4% of a typical retailer's sales (Douthit, Flach, & Agarwal, 2011). Stock, Speh, and Shear (2006) report that OEMs spend 2–3 times more on the cost of processing a return than they spend on the original outbound logistics of the same product. For retailers,[i] if returned products are sold to external liquidators and outlet stores, their salvage value is often only 10–20% of the original value. At the macro level, Enright (2003) reports that the retail sector's annual spending on reverse logistics for the processing and disposition of returns is more than $40 billion.

As a result of such volume and cost, companies often invest in methods to reduce the cost of providing generous return policies, which includes both passive and active strategies. Passive strategies include better forecasting of returns and the subsequent operational adjustments (Toktay, van der Laan, & Brito, 2003). Active strategies, on the other hand, involve searching for ways to reduce returns, which has drawn increasing attention from retailers in recent years (Röllecke, Huchzermeier, & Schröder, 2018). One well-researched active strategy is to impose restocking fees on returns, yet such fees are rarely implemented in practice (Shang, Pekgün, Ferguson, & Galbreth, 2017b). Other than restocking fees, two additional types of active strategies often practiced by retailers are *adjusting the return time window* and offering additional *buyer assistance* during the store visits. Providing empirically grounded guidance regarding how to better manage these two active

[i] The party responsible for a returned product is typically negotiated as part of the OEM's contract with the retailer, with lower wholesale prices offered if the retailer assumes the financial cost of product returns.

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strategies is our primary focus in this article. We introduce each of these strategies briefly below.

First, consider the retailer strategy of adjusting the return time window. Well-publicized examples include Costco, which ended its lifetime return policy on consumer electronics (Hudson, 2007), and REI's decision to shorten its return time window from unlimited to one year (Grind, 2013). Shortening a return time window can "chop off" the long tail of the return age, which usually contains the most costly returns. Furthermore, customizing the return time windows across product categories might also enable a retailer to take advantage of the heterogeneous time that consumers take to evaluate different products. However, such heterogeneity must be measured before it can be managed effectively. Our research provides the first empirical measurement, to our knowledge, of this evaluation time frame across product categories.

A second active strategy that retailers employ is buyer assistance programs, which are designed to help buyers to resolve some of the uncertainty about products and find the best match[ii] (Xia, Xiao, & Zhang, 2017). Typical buyer assistance practices include floor samples, detailed product descriptions, staff–consumer interaction (Ferguson, Guide, & Souza, 2006; Ofek, Katona, & Sarvary, 2011), or structured, itemized product comparisons (Lawton, 2008). Buyer assistance programs can be costly, however, and few stores have the resources to provide sufficient coverage for all product categories. Thus, our research helps retailers focus their assistance efforts by empirically assessing which product categories, and which specific products within a category, are most likely to be returned.

While field experiments offer a clean way to demonstrate the effectiveness of active return reduction strategies, they are often very expensive to implement. Thus, in this article, we present an alternative for retailers—a model that utilizes data collected through standard point-of-sale (POS) technologies to glean insights into where our active practices of interest (buyer assistance and time window adjustment) are likely to be most effective.

In order to target buyer assistance, retailers must understand where consumer uncertainty is highest, and thus returns are most likely to happen. In this article, we hypothesize three key drivers of a consumer's likelihood to return a product. First, where a product stands in its life-cycle (product maturity) determines how familiar consumers are with the product as well as the amount of public information available (user reviews, etc.), both of which can help reduce uncertainty. Second, more competing products can help consumers locate the most suitable product to meet their needs (leading to lower uncertainty about the purchase), reducing return probability. Third, we hypothesize a positive association between a consumer's previous return activity and future returns. This is interesting given the extant literature linking returns to higher future *purchases* (Petersen & Kumar, 2009; Griffis et al., 2012), which implies that generous returns policies have longer term benefits. If previous returns are also linked to future *returns* as well as future sales, this would imply that the net demand/profit impact of returns is more nuanced than has been previously proposed.

[ii] Defective returns only make up 5% of all returns (Lawton, 2008). Product returns due to other reasons, such as difficulty in installing or using the product, unclear descriptions of product features, or poor labeling, can be reduced through buyer assistance efforts at the retailer.

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Table 1: Key analytical studies on optimal restocking fee.

	Key Modeling Elements
Shulman, Coughlan, and Savaskan (2009)	Monopolist, multiproduct
Shulman, Coughlan, and Savaskan (2011)	Horizontal duopoly with exchanges
Shang, Ghosh, and Galbreth (2017a)	Opportunistic consumers
Ketzenberg and Zuidwijk (2009)	Joint order quantity and restocking fee decision
Akçay, Boyacı, and Zhang (2013)	Joint order quantity and restocking fee with open box sales
Hsiao and Chen (2012)	Quality risk
McWilliams (2012)	Vertical duopoly
Hsiao and Chen (2014)	Explicit restocking fee versus a hassle cost
Su (2009); Crocker and Letizia (2014)	Supply chain contracting
Hsiao and Chen (2015)	Decision to refuse returns
Altug and Aydinliyim (2016)	Strategic consumers and markdown pricing
Alptekinoğlu and Grasas (2014)	Assortment planning with returns

Beyond estimating the likelihood of return, we are also interested in how long the consumer takes to make the keep/return decision. This is an underexplored area within the consumer returns literature, which tends to focus on the consumer's decision of "whether to return," rather than "when to return." The lack of literature in this area makes building hypotheses around the time dimension of returns difficult. Given its relevance to practice, however, we provide an exploratory analysis of this aspect of returns in Section 6.

We test our hypotheses using a publicly available dataset provided by a major U.S. consumer electronics retailer (Ni, Neslin, & Sun, 2012) containing 69,782 transactions with 8,152 returns. Product categories include TV, computer, audio speakers, and others. Our results provide empirically grounded insights into the impact of product life-cycle, product variety, and prior returns behavior on return probability. We also comment on the heterogeneity of experience duration, finding interesting insights in terms of product categories as well as seasonality. Finally, our model enables us to predict the age profile of returns, which can help retailers in their allocation of resources across different recovery options.

LITERATURE REVIEW

Our research contributes to the rapidly growing literature on consumer returns. Most articles in this domain, like our study, take a retailer's perspective. Related to our focus on active strategies to manage returns, there are numerous analytical studies addressing how to optimize the restocking fee in conjunction with other decision variables (Table 1). As we elaborate in Section 3.2, some of the assumptions in these articles motivate our hypotheses on the relationship between product variety and a customer's tendency to return. Overall,

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the main focus of this stream of theoretical work has been on the restocking fee as a lever to influence consumer returns (with most articles finding that a positive restocking fee is optimal). We complement these studies by providing guidance to retailers for the less well-studied active strategies of buyer assistance and return time window adjustment. The only analytical study we are aware of that mentions buyer assistance programs is Ferguson et al. (2006), who show how an OEM can use a target rebate contract to induce its retailer to actively reduce returns under the assumption that the retailer can influence return rates through buyer assistance programs. Our study complements this work by identifying product attributes where the retailer can better focus her buyer assistance efforts.

The empirical literature on consumer returns is much smaller than the analytical literature, and it can be organized around two primary themes—*antecedents* of consumer returns and *positive impacts* of consumer returns (Table 2). We contribute to this empirical literature by hypothesizing and testing three important drivers of return probability that are absent in prior studies—product maturity, product variety, and past return experience. While all three of these drivers inform our understanding of the antecedents of returns (and thus the potential focus of buyer assistance programs), the return experience effect also urges a more holistic view of the potential positive impact of return-prone customers. Specifically, we show that while return-prone consumers tend to buy more, they are also more likely to return products, and they take longer to do so than nonreturn-prone consumers.

Despite the fact that the time consumers take to decide whether to keep their purchase is an important consideration for retailers, we find very limited research that touches on this dimension of consumer returns. In an early study in this area, Hess and Mayhew (1997) show that the price of a product does not have an impact on the timing of return. Furthermore, one experimental result from Wood (2001) is that a more lenient return policy does not encourage or discourage consumers to spend more time to evaluate their product. Heiman, McWilliams, Zhao, and Zilberman (2002) incorporate consumers' experience duration into their analytical model and show a positive value of a longer return time window. Some preliminary empirical evidence of this positive value is provided in Heiman, Just, McWilliams, and Zilberman (2015) through a survey instrument and in Rao et al. (2018) through a field experiment on eBay. Other analytical work in this area has studied the coordination between restocking fee and return time window in combating fraudulent returns (Ülkü, Dailey, & Yayla-Küllü, 2013) or how to incorporate the return time window as a strategic decision for a fashion retailer who "circulates" returns in a closed-loop supply chain (Difrancesco, Huchzermeier, & Schröder, 2017). We contribute to this literature by providing a more systematic examination, based on secondary POS data, of whether our postulated drivers of return probability also affect the length of post-purchase product trial. Our exploratory analysis provides the first data-driven evidence of heterogeneity in consumer experience duration across products and seasons, which can be useful to retailers when determining appropriate return time windows.

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Table 2: Key related empirical studies on consumer returns.

	Context	Data Type	Findings
Rao, Rabinovich, and Raju (2014)	Online fashion items	Transaction	Product scarcity information and unreliable delivery performance increase returns.
Anderson et al. (2009)	Mail-order apparel	Choice Incident	Within-product price increases induce returns.
Hess and Mayhew (1997)	Mail-order apparel	Transaction	Higher priced products are more likely to be returned.
Petersen and Kumar (2009)	Mail-order Multicategory	Transaction	Products purchased as gifts, during the holiday season, through a new channel, and on sale are more likely to be returned. A higher dollar value of current return is correlated with higher value of future purchase.
Griffis et al. (2012)	Online educational Material	Transaction	Consumers who returned in the past tend to order more frequently, include more items in an order, and buy more expensive items than those who never returned.
Mollenkopf, Rabinovich, Laseter, and Boyer (2007)	Multiple e-retailers	Survey	Both perceived value of returns offering and return satisfaction increase customer long-term loyalty.
Ertekin (2018)	B&M jewelry	Transaction	Convincing a customer to exchange a returned product will induce more future purchases only if she is satisfied with the replacement as well as the return experience.

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HYPOTHESIS DEVELOPMENT

Retailers allow consumer returns because the true value of a product closely depends on the fit of its attributes with a buyer's needs, and for items such as consumer electronics, the assessment of this fit takes significant consumer effort (Nelson, 1970). Faced with this uncertainty regarding fit, the consumer first estimates how much utility she expects to obtain from the product when making a purchase. After a trial period, the uncertainty is resolved and she adjusts her estimate to her true utility. If the adjustment in utility drops the realization to a point below the price she paid (minus the hassle cost of returning the product), she returns the product. Otherwise, she keeps it. The range of this utility adjustment is commonly referred to as "trial uncertainty" in the literature (e.g., Su, 2009; Shang et al., 2017a). Both a low *ex ante* valuation of the product (relative to other alternatives) and a large trial uncertainty result in a higher probability of a return, *ceteris paribus*. In the following, we hypothesize how a consumer's decision to return can be explained by the product attributes of maturity and variety, as well as the consumer attribute of past return behavior.

Product Maturity

As discussed in Nelson (1970), all products can be considered as being located somewhere in the search-experience spectrum. A particular product's position on the spectrum depends on the consumer's sources of information. If external sources such as advertisement, expert opinions, and word of mouth already provide candid and detailed descriptions about the product, consumers can *search* this publicly available information. Sometimes, the value of a product can be accurately assessed solely through such search. In other words, when a product is closer to the search end of the spectrum, a consumer's uncertainty at the point of purchase is lower. On the contrary, if public information regarding a product is relatively scarce, or if the value of the product is highly personal and idiosyncratic, consumers must rely more heavily on their own *experience* to determine its value. That is, it requires some hands-on trial experience for them to fully assess the product. In other words, lack of knowledge on the "experience attributes" of a product makes it difficult for consumers to resolve fit uncertainty before a purchase (Rao & Monroe, 1988).

One prominent factor that can shift a product's position on the search-experience spectrum is the length of the product's existence in the marketplace. For example, consider a newly introduced television model versus one that has been available for six months. The amount of information (e.g., expert reviews, consumer reviews, defective rates, and performance data) available to the consumer is much more abundant for the latter (Kalish, 1985). This observation coincides with how word-of-mouth effects are incorporated into product diffusion models (Mahajan, Muller, & Bass, 1990). The fact that publicly available product information accumulates over time is more relevant than ever due to the popularization of consumer forums and online recommendation systems (Dimoka, Hong, & Pavlou, 2012). In general, as a product becomes more mature in the market, more information is disclosed through various sources (Ghosh & Galbreth, 2013) and hence its value can be more precisely gauged by consumers without hands-on experience. Therefore, consumers' trial uncertainty of a more mature product is expected to be

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lower, which in turn leads to a lower probability of a product return. Accordingly, we make the following hypothesis:[iii]

Hypothesis 1: Product maturity is negatively related to return probability.

Product Variety

The number of options available at a retailer within a particular product assortment (i.e., product variety) can also influence the likelihood of a consumer return. Prior literature streams from marketing, psychology, and economics have provided comprehensive insights on how product variety or assortment size affects consumers' purchase decision process and post-purchase satisfaction (see McAlister and Pessemier (1982); Mantrala et al. (2009) for a survey). Using these insights, together with the unique characteristics of the return context, we hypothesize an *uncertainty reduction mechanism*, through which variety exerts its impact. Hong and Pavlou (2014) categorize product uncertainty into fit and quality uncertainty and show that it is the fit uncertainty that drives returns. Casting heterogeneous consumers on a spectrum of product preference, fit uncertainty is manifested such that a consumer is not exactly sure where her location on the spectrum is. In fact, this is exactly how fit uncertainty is modeled in analytical studies of consumer returns, such as in Shulman et al. (2011). In consumer behavior literature, the process of searching one's exact preference is called "ideal point articulation" (Chernev, 2003a, 2003b). When faced with an assortment of product choices, comparing the design attributes of these alternatives provides an effective way to articulate the ideal point (Brickman & D'amato, 1975). As variety increases, consumers are likely to make more detailed between-product comparisons, reducing fit uncertainty. In addition, Huffman and Kahn (1998) demonstrate that with a more articulated ideal point, consumers tend to be more satisfied after the purchase. Therefore, through the uncertainty reduction mechanism, more variety increases the confidence that the best choice has been made, which in turn reduces the return probability.[iv] As a result, we make the following hypothesis:

Hypothesis 2: Product variety is negatively related to return probability.

After returning a product, consumers might follow up with an exchange or give up purchasing from this retailer altogether (Ertekin, 2018). We carry out additional analysis to distinguish the impact of product variety on exchanges versus pure returns.

Return Experience of the Consumer

Our last hypothesis focuses on an attribute of the consumer rather than the product. It has been shown in previous studies that both the dollar amount and the frequency

[iii] Note that one might argue that there is also a positive relationship between maturity and return probability, arising from more substitutes as a product matures. However, because we also hypothesize variety as a return driver and measure it by the number of competing alternatives, we believe this alternative maturity–return relationship is controlled for in our analysis. We thank an anonymous reviewer for bringing this to our attention.

[iv] One might argue that there is an opposite *differentiation reduction mechanism* that increases return probability. That is, higher variety makes the chosen product more similar with the second best choice, which increases the appeal of the exchange option in the consumer's return decision. Our empirical context is consumer electronics, which have relatively lower variety compared with other products that are susceptible to returns such as apparel and jewelry. Thus, *differentiation reduction* is less likely to be at work.

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of future purchases positively correlate with a consumer's past return behavior. In other words, people who return more frequently tend to make more purchases and for higher dollar amounts than do consumers who return less frequently. In the context of online book retailing, Griffis et al. (2012) construct a quasi-experiment to show that consumers who have previously returned a product make more frequent orders and purchase more expensive items compared with the consumers who have not returned products in the past. In addition, Petersen and Kumar (2009) empirically estimate an inverted U-shaped relationship between the amount of returns a consumer has made in the past and the value of that consumer's future purchases, again indicating a positive correlation (at least to a point). In the same spirit, Mollenkopf et al. (2007) provide evidence that consumers with satisfactory return experiences exhibit higher loyalty to the retailer than do customers who have not returned a product. Absent in this previous research, however, is any analysis to determine whether consumers who made more returns in the past have a higher likelihood of *returning* products in the future. Given the high cost of processing returns (Stock et al., 2006), from a retailer's perspective the profitability of a particular consumer can be estimated only when information is available on both the future demand and the future return rate.

In considering why return experience might impact future return behavior, we use the classification of buyer–seller interaction costs from transaction cost theory (Williamson, 1985; Chircu & Mahajan, 2006) of time-type costs and psychological-type costs. In the context of product returns, psychological-type costs could include the embarrassment or anxiety associated with returning a product, while time-type costs may be associated with the time and inconvenience of making a return. Based on these concepts, we suggest that consumers with more return activity may have lower psychological-type costs (e.g., they are less embarrassed by making a return) and/or lower time-type costs (e.g., they live closer to the retailer location, or they inherently value their time less). Thus, from either the psychological-type or the time-type perspective, we expect prior return experiences to predict higher future return probabilities:

Hypothesis 3: Consumers' past return experiences are positively correlated with their future return probability.

DATA AND VARIABLES

As an effort to promote quality empirical research, the Institute for Operations Research and the Management Sciences (INFORMS) encourages scholars to publish unique and comprehensive datasets. The outcomes of this effort include three datasets published in *Manufacturing & Service Operations Management* (Willems, 2008; Bodea, Ferguson, & Garrow, 2009; Mumbower & Garrow, 2014) and four in *Marketing Science* (Bronnenberg, Kruger, & Mela, 2008; Goldenberg, Libai, Muller, & Stremersch, 2010; Ni et al., 2012; Wang, Mai, & Chiang, 2014). We make use of the Ni et al. (2012) dataset for the following reasons. First, the National Retail Federation (2018) estimates that return rates for "hard goods," including consumer electronics, are 50% more than the average across product categories, highlighting the importance of developing effective return reduction programs.

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Second, Ni et al. (2012) suggest that their data are by far the most comprehensive dataset of consumer electronics transactions available. Third, researchers are specifically encouraged to work on product return issues with these data (Ni et al., 2012, p. 1012). Fourth, given the semipublic nature of the data, our results are easily replicated and can be used as building blocks for future research. The original data contain 173, 262 transactions provided by a major U.S. electronics chain, the identity of which is not provided for confidentiality reasons. All product returns at this retailer are accepted open-box and given a full refund.[v] The transactions in this dataset took place between December 1998 and November 2004. We limit our attention to a subset of the original data that is applicable to our analysis. A detailed description of how we obtained our final dataset is provided in online Appendix A.

The applicable data for our analysis contain 69, 782 transactions spanning across nine product categories. Return lag is calculated by differencing the purchase and return timestamps. Table 3 provides descriptive statistics for the return rate and lag of each category. The range of return rates observed in our data matches closely with the 12% average for consumer electronics reported by the National Retail Federation (2014). The average return lag ranges from 11 to 28 days, which motivates a more refined return time window policy than the common single time window for products.

As is typical with retail transaction data, product exchanges are not automatically identified in our dataset. In other words, we need to match an exchange, if there is one, to an earlier return made by the consumer. Specifically, we define exchange in two ways—a more stringent same-day exchange and a more liberal 7-day return. If a consumer purchased another product in the same subcategory as the returned item on the same day (within 7 days), then we code this observation as a same-day exchange (7-day exchange). Examples of these two types of return-exchange matching are given in online Appendix A (Table OA.2). Note that same-day exchange is a subset of the 7-day exchange and hence the latter is a more liberal definition of exchanges. As shown in Table 3, exchange rate is between 1% and 5% in our sample. In general, they make up less than one third of all returns. Next, we discuss the construction of our independent variables. Table 4 lists the descriptive statistics.

Maturity is measured by differencing the dates of the focal transaction and the first transaction for the same product. For example, assume that 12 transactions are recorded for a certain digital camera. If the fifth transaction was recorded on 07/06/2012 and the first on 02/22/2012, maturity is recorded for the fifth transaction as 134 days.

Variety is operationalized by counting the number of competing products. The retailer has a detailed product categorization scheme, which we use to define competing products. Within big categories such as TVs and computers, products are further grouped by their size and features. For example, while 9"–16" color TVs belong to one subcategory, 19"–20" color TVs belong to another. We measure

[v] We do not observe the reason for return in our data. To our knowledge, most retailers either do not collect these data due to the "no question asked" policy or have a poor accuracy when recording it. Thus, we cannot tease out the defective returns. However, these returns make up only 5% for a typical retailer (Lawton, 2008; Douthit et al., 2011).

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Table 3: By-category descriptive statistics for returns, exchanges, and return lags.

		Exchange		Return Lag			
	Return	Same day	Within 7 days	Mean	Std. Dev.	95th Percentile	Observations
Audio	11.72%	2.46%	2.88%	13.44	20.63	44	9,595
Auto parts	14.08%	4.57%	5.05%	22.23	57.01	105	5,782
Cable	11.89%	2.13%	2.41%	10.95	20.73	42.5	6,394
Computer	10.70%	2.57%	2.96%	11.51	23.78	46	18,429
Imaging	10.59%	2.27%	2.58%	12.51	21.22	43.5	6,041
Mobile phone	16.07%	3.63%	4.51%	28.12	79.31	109	2,371
Phone	14.43%	4.23%	4.73%	16.95	32.31	65	4,921
TV	8.17%	1.50%	1.77%	11.79	22.59	41	8,385
TV box	13.54%	3.29%	3.73%	15.17	28.56	57	7,864

"Auto parts" are off-market speaker, video, and GPS systems for cars. "TV box" are recorder, DVR, VCR, DVD player, and other box-shaped items that connect to TV.

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Table 4: Variable descriptive statistics.

	Description	N	Mean	Std. Dev.	Min.	Max.
Dependent Variables						
Return	Dummy variable with 1 denoting return	69782	0.12	0.32	0	1
Exchange	Dummy variable with 1 denoting same-day exchange	8152	0.24	0.42	0	1
*Exchange 7 day**	Dummy variable with 1 denoting seven-day exchange	8152	0.27	0.44	0	1
Return Lag	Days from purchase to return	8152	14.63	33.88	0	994
Focal Independent Variables						
Maturity	Days from product introduction to purchase date	69782	172.80	208.93	0	2041
Variety	# of competing products in the same assortment on purchase date	69782	20.13	16.39	1	119
*Variety Return day**	# of competing products in the same assortment on return date	8152	18.36	15.13	1	119
Return Exp	# of returns made by the customer before the purchase date	69782	0.53	1.36	0	19
*Return Exp 6 Month**	# of returns made by the customer within past six months	69782	0.15	0.58	0	14
Control Variables						
Price Ratio	(purchase price − average product price)/average product price	69782	0	0.12	−0.79	0.79
Average Price	Average product price	69782	227.64	342.48	5.04	5364.99
Discount	Dummy variable with 1 denoting sold on discount	69782	0.01	0.1	0.0	1.0

Continued

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Table 4: Continued

	Description	N	Mean	Std. Dev.	Min.	Max.
Length	Days from product introduction to date of last purchase	69782	347.62	293.97	1	2042
Past Avg Return Cat[*]	Average return rate for the assortment in the past month	68661	0.13	0.12	0	1
Popularity	Average number of the product's monthly sales (standardized)	69782	0.04	0.87	−0.56	13.41
Basket Size	# of items in the customer's basket	69782	2.98	2.22	1	17
Basket Value	Total $ value of the basket	69782	586.55	772.12	2.20	12265.91
Past Purchase	Total $ amount of purchases in the past six months for the customer	69782	549.63	880.93	0	23499.78
Visit Frequency	Customer's average monthly visits to the store in the past six months	69782	0.11	0.22	0	3.7
Holiday	Dummy variable with 1 denoting November and December	69782	0.21	0.41	0	1
Age[*]	Customer's age	60905	48.56	14.39	18	99
Gender[*]	Dummy variable with 1 denoting male	63583	0.34	0.47	0	1
Children[*]	# of children in the customer's household	69782	0.46	0.85	0	6

[*]These variables are used in robustness checks.

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the number of competing products by counting all other products in the same subcategory in the month of the *focal purchase* transaction. For instance, to compute the number of competing products for a 19" TV purchased in March of 2003, we count the number of other transacted products that belong to the 19"–20" color TV subcategory in the same month.[vi] As an alternative measure, *VarietyReturnDay* counts all other products in the same subcategory in the month of the *focal return* transaction. We use *VarietyReturnDay* in the robustness checks for the variety–exchange relationship, because consumers might check the assortment on the return date before making an exchange decision.

Whenever a consumer makes a purchase, we count the number of products that consumer has returned before the transaction date. Thus, *ReturnExp* is operationalized as the number of past returns.[vii] We also count the number of returns in the past six months for a focal purchase and label it as *ReturnExp6Month*, which is used as an alternative measure in the robustness check.

Because we pool an array of products together for analysis, we control for the price of a transaction through three variables, *AveragePrice*, *PriceRatio,* and *Discount*. *AveragePrice* is the within-product average of transaction prices net any discount. *PriceRatio* is calculated as $\frac{\text{purchase price}-\text{average price}}{\text{average price}}$, which captures the fluctuation of price within products and is comparable across products. The binary variable, *Discount*, captures whether a product is sold with a discount.

The popularity of a product might be correlated with return probability. To make it comparable across product categories, we operationalize *Popularity* as the average number of a product's monthly transactions, standardized within product category. *Length* takes the difference between the dates of last and first purchases of a product, which approximates the number of days a product has been in existence with the retailer. Controlling for *Length* reduces a potential endogeneity concern in the maturity estimates such that longer maturity might be due to longer lifespan of a product, which in turn might have better quality and hence lower return probability. *PastAvgReturnCat* measures the average return rate for the assortment (i.e., subcategory) in the past month, which is used in a robustness check to alleviate endogeneity concern on *Variety* arising from return-based assortment adjustment.

We control for the characteristics of a consumer's basket through two variables, *BasketSize* and *BasketValue*. The former counts the number of items in the consumer's basket when the focal transaction was completed. The latter is the total dollar value of the basket. The number of past returns of a consumer is correlated with her purchase pattern and store visit frequency. To remove potential omitted variable bias introduced by these consumer characteristics,[viii] we measure *PastPurchase* by the total dollar amount of purchases in the past six months for a consumer and *VisitFrequency* by the average number of monthly visits to the store in the past six months. The latter helps alleviate endogeneity concerns on *ReturnExp*, which

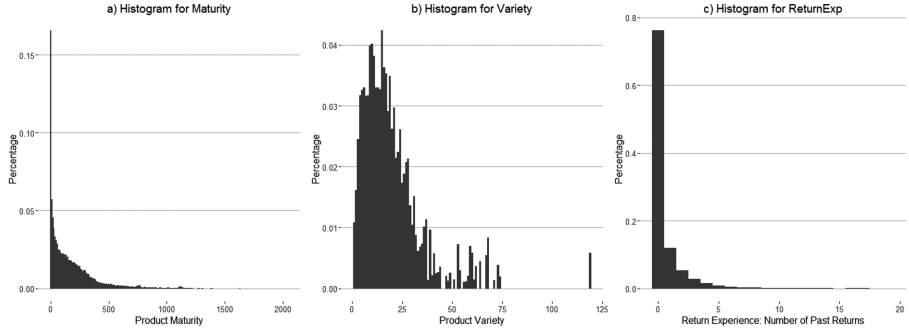
[vi] We have verified that all products have at least one transaction per month between the first and last transactions.

[vii] If a consumer makes a return and then a new purchase on day *t*, this return will not be added to the return experience variable for the observation that represents the same-day purchase. However, for any purchases this consumer makes after day *t*, the above return will be counted in the return experience variable. We also note that the return experience variables calculated here will not capture returns made to other retailers. We thank an anonymous reviewer for pointing out this limitation of our research.

[viii] For example, frequent visitors are expected to have lower opportunity cost of making a return trip, which leads to a higher chance of a return.

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Figure 1: Histograms.



may arise from omitted factors such as the cost of making a return trip. Because we cannot directly observe this cost in the data, *VisitFrequency* serves as a proxy. Three additional consumer demographic variables, *Age*, *Gender*, and *Children*, are used in robustness checks.

Given our focus on consumer electronics, peak sales occur during the November and December holiday season. According to the National Retail Federation (2014), the return rate in the holiday season is 11% higher than regular times, which can be attributed to gift and impulsive purchases. For example, Petersen and Kumar (2009) argue that consumers may purchase more products that either do not fit their needs or they may find unnecessary later on. We create a dummy variable *Holiday*, which is 1 for transactions made in November and December. Finally, we also use categorical dummy variables to control for the difference across categories.

RETURN PROBABILITY ANALYSIS

We begin our analysis by observing the histograms of our focal independent variables—maturity, variety, and return experience (Figure 1). Maturity has strong spikes at small values, which is expected because all products have transactions with small maturity but only those with longer product lives have larger maturity values. However, the long tail raises the concern that any estimated effects will be driven by the larger values (i.e., those on the tail) and are thus spurious. Thus, we log transform *Maturity* by log(*Maturity*+1). For variety, we observe a number of potential outliers with relatively large values. This concern is addressed in our robustness checks by excluding products with large variety values. Last, a large proportion of the customer observations have no past returns, which is due to the fact that many consumers did not make a return in the entire time span of the data.[ix] In a robustness check presented in online Appendix B, we test the difference between return and nonreturn consumers.

[ix] Note that although the shape of the return experience histogram also resembles an exponential distribution, it has less of a long tail problem as in the maturity case. Results are qualitatively the same if we replace return experience with its log-transformed version.

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Regressions

We test our hypotheses by fitting a Probit regression in the following form through Stata 14's `probit` command:

$$Return_i = \begin{cases} 1 & \text{if } Z_i\Gamma + u_i > 0, \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where u_i is a standard normal random variable, Z_i is the vector of independent variables, and Γ is the vector of coefficients. Results are provided in Table 5. Model 1 includes only control variables. Model 2 adds the three hypothesized variables into the regression. The following shows the specification of $Z_i\Gamma$ in model 2:

$$\begin{aligned} Z_i\Gamma = {} & \gamma_0 + \gamma_1 Log\,Maturity_i + \gamma_2 Variety_i + \gamma_3 Return\,Exp_i + \gamma_4 Price\,Ratio_i \\ & + \gamma_5 Average\,Price_i + \gamma_6 Discount_i + \gamma_7 Length_i + \gamma_8 \text{Popularity}_i \\ & + \gamma_9 \text{Basket Size}_i + \gamma_{10} \text{Basket Value}_i + \gamma_{11} \text{Past Purchase}_i \\ & + \gamma_{12} \text{Visit Frequency}_i + \gamma_{13} \text{Holiday}_i + \Sigma_{j=14}^{22} \gamma_j d_{ij}, \end{aligned} \quad (2)$$

where d_{ij} are the categorical dummy variables.

Our results are presented in Table 5. We observe a negative effect of maturity on predicting the probability of a return (model 2), supporting H1. That is, compared with a product existing on the shelf for one year, a newly introduced product is associated with a higher return rate. In terms of product variety, we also observe a negative effect, supporting H2. Regarding return experience, H3 is supported—a product purchased by a consumer with a higher number of past returns has a higher likelihood of also being returned. In Section 6, we show that return experience is also related to longer return lags, adding to a retailer's cost of serving the return-prone consumers. The significance levels for the three focal variables ($p < .001$) are replicated by a likelihood ratio test.

Next, we turn to exchanges—model 3 in Table 5. The appropriate modeling approach depends on our assumption about whether the exchange decision is made after the return decision (sequential) or together with the return decision (simultaneous). We believe the sequential assumption is empirically more realistic for two reasons. First, the exchange decision is most likely to be finalized in the store, after the decision has been made to make a return trip. Second, there is a nontrivial fraction (13%) of exchanges made after the return date in our data.[x] Thus, in the following analysis, we present results from a sequential Probit model. However, the simultaneous results based on a Multinomial Probit model were also performed as a robustness check and the results showed a strong consistency with the sequential method.

The two-stage sequential Probit model fits two Probit models, with a first stage for the return versus keep decision and a second stage for the exchange versus no exchange decision (among the returns). While the first stage is essentially the same as model (2), the specification for the second stage is as follows:

$$Exchange_i = \begin{cases} 1 & \text{if } Z_i A + e_i > 0, \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

[x] Recall that we have calculated two exchange measures, *Exchange* and *Exchange7Day*. Exchanges defined by the former make up 87% of exchanges defined by the latter.

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Table 5: Probit regressions for the effects of product maturity, variety, and return experience.

	Model 1		Model 2		Model 3	
Dependent Variable	Return		Return		Exchange	
Log Maturity			**−0.011**	$(-2.92)^{**}$	0.034	$(3.36)^{***}$
Variety			**−0.0050**	$(-9.15)^{***}$	**0.0090**	$(7.56)^{***}$
Return Exp			**0.059**	$(12.83)^{***}$	−0.0035	(−0.32)
Price Ratio	0.10	$(1.86)^{\dagger}$	0.081	(1.45)	−0.11	(−0.84)
Average Price	-0.13×10^{-3}	$(-4.68)^{***}$	-0.14×10^{-3}	$(-5.07)^{***}$	-0.18×10^{-3}	$(-2.39)^{*}$
Discount	−0.68	$(-7.71)^{***}$	−0.70	$(-7.83)^{***}$	−0.24	(−0.78)
Length	−0.046	$(-5.90)^{***}$	−0.036	$(-4.07)^{***}$	−0.068	$(-3.42)^{***}$
Popularity	−0.054	$(-5.15)^{***}$	−0.028	$(-2.71)^{**}$	−0.062	$(-2.40)^{*}$
Basket Size	0.041	$(11.04)^{***}$	0.041	$(10.92)^{***}$	0.12	$(13.37)^{***}$
Basket Value	0.19×10^{-3}	$(11.64)^{***}$	0.20×10^{-3}	$(11.82)^{***}$	-0.28×10^{-4}	(−0.98)
Past Purchase	-0.17×10^{-3}	$(-10.63)^{***}$	-0.19×10^{-3}	$(-11.60)^{***}$	-0.78×10^{-4}	$(-2.72)^{**}$
Visit Frequency	0.27	$(7.42)^{***}$	0.13	$(3.28)^{**}$	0.14	(1.51)
Holiday	0.16	$(9.63)^{***}$	0.15	$(9.20)^{***}$	0.0046	(0.13)
Categorical fixed effects	included		included		included	
Constant	−1.11	$(-23.65)^{***}$	−1.056	$(-21.81)^{***}$	−0.98	$(-8.98)^{***}$
Sample size	69,782		69,782		8152	
Logged likelihood	−24,559.40		−24,403.40		−4241.00	

$^{\dagger}p < .1, ^{*}p < .05, ^{**}p < .01, ^{***}p < .001$. t-statistics in parenthesis, calculated by cluster robust standard errors with product as the grouping variable.

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where e_i is a standard normal random variable, Z_i is the identical vector of independent variables as in Equation (2), and A is the vector of coefficients. Model (3) shows that more variety is associated with a higher chance of exchange among the returned items. Together with the negative variety effect in model (2), we conclude that more variety is indeed associated with fewer pure returns. This result is consistent with the theory that only the uncertainty reduction effect of variety is at work for returns without exchange. Because the total amount of returns decreases with variety yet the fraction of exchanges within it increases, the directionality of the variety–exchange link is not clear. We note that, due to potential sample selection bias, our results from the exchange regression should be interpreted only for the returned purchases, but not to be generalized to those that are kept.

Due to the nonlinear nature of the probit regression, the coefficients do not carry the same marginal effects interpretation as in a linear multiple regression. Therefore, in the next section, we calculate two sets of average marginal effects (AMEs) which demonstrate the economic significance of our results and provide a straightforward interpretation of the variety–exchange link.

Average Marginal Effects

The first set of AMEs numerically illustrates the impact of maturity and variety on the return probability (including both exchange and pure return), given their close relevance to buyer assistance programs. The left part of Table 6 shows point estimates of the return probability evaluated at various representative values of *Maturity* and *Variety* along with the AMEs.[xi] The effect of maturity is much larger when the products being compared are early in the life-cycle, such as new versus 6 months old. Regarding product variety, we observe a relatively constant effect size. Specifically, when the number of competing products increases from 5 to 15, the decrease in the return probability is similar to what is seen at increases from 15 to 25.

The second set of AMEs focuses on the variety–exchange link. Specifically, we compute how the probabilities of exchange[xii] and pure return change as variety increases from 5 to 15, and then to 25. Results are presented in the right side of Table 6. We make two observations. First, variety only slightly increases the exchange probability, which is statistically insignificant. Second, more variety considerably reduces pure returns, the magnitude of which is even larger than its impact on total returns. Overall, our variety results send a practical message to retailers: more product variety significantly reduces the most costly pure returns, while only slightly increasing exchanges (which are much less costly).

Control Variable Effects

We include many controls in our models. First, we look at the price effects. *Discount* has a negative effect on the return probability, replicating the results in Anderson

[xi] We use *Variety* to illustrate this procedure. Assume that we want to know the amount of decrease in the return probability comparing a product with *Variety* = 5 (few competing alternatives) versus 15 (many competing alternatives). For transaction i, $\Pr(Return|Variety_i = 5) - \Pr(Return|Variety_i = 15)$ is the quantity of interest but this quantity is heterogeneous across the sample. Thus, we compute its sample average as the AME of variety, based on model (2).

[xii] For example, $\Pr(Exchange|Variety_i = 5) = \Pr(Return|Variety_i = 5) \times \Pr(Exchange|Variety_i = 5, Return)$. The two right-hand side probabilities are calculated from models (2) and (3), respectively.

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Table 6: Average marginal effects for product maturity and variety.

Outcome		Probability	AME	Outcome		Probability	AME
Return	Maturity = 1	12.57%		Exchange	Variety = 5	2.61%	
	180	11.41%	$-1.16\%^{***}$		15	2.72%	0.11%
	360	11.26%	$-0.15\%^{*}$		25	2.81%	0.09%
Return	Variety = 5	13.13%		No exchange	Variety = 5	10.53%	
	15	12.13%	$-1.00\%^{***}$	or pure return	15	9.42%	$-1.11\%^{***}$
	25	11.19%	$-0.94\%^{***}$		25	8.38%	$-1.04\%^{***}$

$^{\dagger}p < .1$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. AMEs are calculated as the probability in the current row minus that in the upper row.

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et al. (2009). *PriceRatio* captures price fluctuations within a product's sales over time. The analytical research has taken different approaches to model the impact of price on the return rate. When consumers are assumed to be homogeneous, price positively correlates with the return probability because an increase in price lowers everyone's *ex ante* utility of the product. For heterogenous consumers, however, the directionality of the price–return link is confounded. Consumers who have higher product evaluations might find a closer fit with the product as in Shulman et al. (2011), or could perceive a higher hassle cost for returning a product as in Hsiao and Chen (2012), both leading to a lower probability of a return. Our analysis shows that the utility effect of price appears to counteract the higher price effect, which leads to an insignificant overall effect on the return rate. *AveragePrice* allows us to compare products of different price levels. We observe that more expensive products are less likely to be returned, a possible explanation for which is that consumers tend to spend more time searching for the most suitable product for higher priced product categories than they do for lower priced product categories.

Second, we look at popularity effects. When examining the impact of allowing for returns on the optimal assortment (Alptekinoğlu & Grasas, 2014, p. 881), assume that more popular products are less likely to be returned, which is a key assumption that drives their results. We report empirical evidence in support of this assumption.

Finally, we look at holiday period effects. For consumer electronics, peak sales occur during the holiday season. According to the National Retail Federation (2014), the return rate during the holiday season is 11% higher than during nonholiday periods. Such an increase is often attributed to gift purchases and impulsive holiday buyers (Petersen & Kumar, 2009). Interestingly, the AME for the variable holiday (computed the same way as in the previous section) in our data is 23.5%, which is much higher than the unconditional estimate from the National Retail Federation. This is probably due to the fact that many consumers are buying big ticket items during the holiday season, which, as we have shown, are less likely to be returned. Thus, the unconditional estimate is much smaller.

We provide extensive robustness checks for our previous results, which are divided into three sets. The first set uses the same model setup as before, yet different sample, variable, or estimator specifications. In the second set, we fit two fixed-effects logistic regressions for return probability to control for any product-level or store-level heterogeneity. The third set considers the alternative simultaneous assumption for the exchange model. These robustness checks, discussed in detail in online Appendix B, show strong consistency with our main results.

EXPERIENCE DURATION ANALYSIS

From a retailer's perspective, knowing the timing of when products will be returned may be as important as knowing the probability of a return. The timing dimension dictates the age and value of the returns, which has implications for the best salvage strategy (Stock et al., 2006). In addition, understanding the time it takes for consumers to fully experience a product is crucial for setting the appropriate time window for the return policy. Despite its importance, little is known about the drivers of consumers' experience duration (i.e., the time taken to decide whether

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or not to keep an item) and how heterogeneous this time to experience is across product categories. In the following, we provide a set of analysis on the time aspect of returns. First, we examine whether the return drivers—those we have hypothesized and controlled for in the previous analysis—are also associated with experience duration. Second, we show that experience duration exhibits a large variation across product categories and has a spike at the holiday period. Motivated by this observation, we derive reference return time windows that illustrate the benefit of return policies customized for different product categories (as opposed to the uniform time window policies common among today's retailers). Third, retailers often have more than one recovery option for consumer returns, such as restocking them as open-box items, sending them back to the OEM, or liquidating them through third parties. The disposition choice depends critically on the age of the returned product, as older products carry lower resale values. Thus, we provide a prediction model for the proportion of returns that will be relatively new (or old).

The key challenge of estimating experience duration is that it is observable in the data only when the purchase is returned. For consumers who decide to keep their purchase, the time taken to make this decision is not known. As a result, estimating the experience duration using only the data from products that were returned may introduce a sample selection bias (Heckman, 1979).[xiii] To illustrate this bias, we estimate and compare two models—an OLS regression based on the subsample of returns and a Heckman's sample selection model based on the full sample. Because the retailer in our data does not impose a time limit on the returns, the return lag has a relatively long tail, as evidenced in the 95th percentiles shown in Table 4. In addition, a histogram of the return times (available upon request) resembles a log-normal distribution. Thus, we use the log-transformed version of the return lag as the dependent variable. This treatment of our time-to-return dependent variable is consistent with the way time-to-recall is analyzed in Hora, Bapuji, and Roth (2011).

The Heckman's sample selection model (implemented by Stata's `heckman` command) simultaneously estimates the experience duration (i.e., the outcome equation) and the return/keep choice (i.e., the selection equation), which is in the form of

$$\log(y_i) = X_i \mathrm{B} + \epsilon_i$$
$$Return_i = \begin{cases} 1 & \text{if } Z_i \Gamma + u_i > 0, \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

where y_i is the return lag, X_i is the vector of independent variables, and B is the vector of coefficients. Notations in the selection equation are the same as in (1), except that *PriceRatio* is removed due to its insignificance as shown in model 2.

[xiii] Certo, Busenbark, Woo, and Semadeni (2016, p. 2643) describe two conditions in order for sample selection bias to exist. First, the independent variables of interest (e.g., *Variety*, *Maturity*, and *ReturnExp*) should be significant predictors in the selection equation (i.e., return probability regression). Second, the error terms of selection and outcome (i.e., experience duration) equations should have a nontrivial correlation. The problem on hand obviously satisfies the first condition. For the second condition, there are reasonable arguments for both a positive and a negative correlation. For example, consumers who regret a purchase might be more likely to return the product and also be more likely to end the product trial early (i.e., a negative correlation). Conversely, a customer who pays keen attention to details might spend a long time testing the product, yet having a high return probability (i.e., a positive correlation).

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Table 7: OLS and Heckman regressions of experience duration.

	Model ED1 OLS		Model ED2 Heckman	
LogMaturity	−0.013	(−1.58)	−0.0084	(−0.97)
Variety	−0.0014	(−1.43)	0.00029	(0.27)
ReturnExp	0.042	(4.37)***	0.024	(2.38)*
Length	0.025	(1.58)	0.032	(2.03)*
BasketSize	0.030	(3.85)***	0.018	(2.16)*
BasketValue	0.72×10^{-4}	(3.96)***	0.72×10^{-4}	(1.80)†
VisitFrequency	−0.29	(−3.55)***	−0.21	(−2.69)**
Holiday	0.35	(10.50)***	0.299	(8.54)***
Categorical fixed effects: base is audio				
Auto parts	−0.013	(−0.18)	−0.055	(−0.76)
Cable	−0.18	(−2.96)**	−0.20	(−3.27)**
Computer	−0.25	(−5.28)***	−0.22	(−4.48)***
Imaging	−0.087	(−1.50)	−0.053	(−0.90)
Mobile phone	0.095	(0.85)	0.011	(0.10)
Phone	0.18	(2.91)**	0.10	(1.64)
TV	−0.24	(−3.90)***	−0.17	(−2.67)**
TV box	0.042	(0.74)	−0.0019	(−0.03)
Constant	1.65	(18.66)***	2.35	(14.85)***
ρ			−0.32	(−5.45)***
Sample size	8,152 (all returns)		69,782 (full sample)	

†$p < .1$, *$p < .05$, **$p < .01$, ***$p < .001$. *t*-statistics in parenthesis. Heckman model simultaneously estimates the outcome and selection equations. Results for the latter are very similar to that in model (2) of Table 5 and are omitted here for simplicity. Cluster robust standard errors with product as the grouping variable. *Discount*, *AveragePrice*, and *Popularity* are not significant in the duration equation if added.

The error terms follow a bivariate normal distribution such that $\epsilon_i, u_i \sim \mathcal{N}(0, \Sigma)$, where $\Sigma = \begin{pmatrix} \sigma^2 & \rho\sigma \\ \rho\sigma & 1 \end{pmatrix}$—the standard setup of sample selection model (Heckman, 1979).[xiv] When fitting Heckman's model, it is recommended to have at least one variable in the selection equation that is not included in the outcome equation. We believe *Discount*, *AveragePrice*, and *Popularity* have nothing to do with the time consumers take to evaluate a product and hence exclude these three variables in the outcome regression. Thus, the specification of $X_i\mathrm{B}$ becomes

$$\begin{aligned} X_i\mathrm{B} = {} & \beta_0 + \beta_1 Log\,Maturity_i + \beta_2 Variety_i + \beta_3 Return\,Exp_i + \beta_4 Length_i \\ & + \beta_5 \mathrm{Basket\,Size}_i + \beta_6 \mathrm{Basket\,Value}_i + \beta_7 \mathrm{Visit\,Frequency}_i \\ & + \beta_8 \mathrm{Holiday}_i + \Sigma_{j=9}^{17} \beta_j d_{ij}. \end{aligned} \tag{5}$$

The OLS model estimates only $\log(y_i) = X_i\mathrm{B} + \epsilon_i$, which is implemented by the `reg` command in Stata. Results from the two models are presented in Table 7.

[xiv] σ is the standard deviation of ϵ_i. Error variance of the selection equation is normalized to 1 for identification purposes. ρ is the correlation between the two error terms.

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The main parameter in the Heckman model that drives the difference of its estimates from the OLS regression is the error correlation, ρ. A negative and significant (at .001 level) correlation means that if an item is more likely to be returned, its experience duration is shorter, conditional on the independent variables. The consumers' idiosyncratic level of buyer remorse and convenience of traveling to a store may be the unobserved factors that cause this negative correlation. A negative ρ also explains why the constant term is noticeably smaller in the OLS estimate—the (conditional) experience time is shorter for the returned items than for the whole sample. This difference is more important when the retailer wishes to estimate the distribution of experience duration for all consumers. One example of such a situation is when the retailer sets the return time window. We have also performed a Hausman's test to examine the systematic differences between the coefficients in the two models (Hausman, 1978), which is significant at the 0.01 level. Therefore, the influence of the sample selection bias has both statistical and practical significance—not accounting for it will cause spurious effects and underestimate the experience duration.

Based on model (ED2), we observe that many drivers of return probability have little impact on the time it takes consumers to experience a product. However, consumers with more past returns generally take longer to experience a product (i.e., a positive coefficient on *ReturnExp*). Combining this finding with our earlier results on return probability, we conclude that these consumers are not only returning more but also returning later. A strong holiday effect suggests that consumers take much longer to make the keep-return decision during the holiday period. Several reasons might explain this: (i) consumers are more likely to be traveling, (ii) customers delay returns in an effort to avoid the holiday crowd, and (iii) some holiday purchases are sent as gifts. In addition, many of our categorical dummy variables are significant, which indicates heterogeneity of experience duration across categories. It appears that larger products require less time to experience. For example, the relative effects of TVs and computers are negative and significant.

Next, consider a retailer that wants to shorten its time window from "no limit" to a certain number of days (such as Costco's decision in 2007). Furthermore, suppose the retailer decides the appropriate window should allow 85% of its consumers enough time to fully experience their products while pushing the tail 15% to accelerate their product evaluation. In this case, the retailer can first estimate the 85th percentile of experience duration for each category-holiday combination and then look for a pattern in these estimates. The estimated percentiles using our data are presented in Table 8 (columns 2 and 3). We interpret these results as follows. First, the 85th percentiles are around 20 days longer during the holiday season, which is in support of a longer holiday return window that some retailers have already implemented. Second, these percentiles tend to form clusters across product categories. For example, during the nonholiday season, cables, computers, and TVs all have their 85th percentile at around 40 days, while audio, auto parts, imaging, and TV box have theirs at around 50 days. Therefore, instead of using a uniform time window, the retailer could consider creating a portfolio of return windows for various product categories. We remark that our empirical results are just one input into the determination of return time windows, which admittedly

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Table 8: Return partitioning and length of experience duration.

	85th Percentile		Return Partitions Based on Newness					
	Experience Duration (Days)		Lag < 7 Days		7 Days < Lag < 30 Days		Lag > 30 Days	
	Nonholiday	Holiday	Nonholiday	Holiday	Nonholiday	Holiday	Nonholiday	Holiday
Audio	49.55	66.90	58.86%	48.08%	31.18%	36.30%	9.96%	15.62%
Auto parts	47.78	64.81	58.52%	47.46%	31.36%	36.52%	10.12%	16.02%
Cable	42.98	58.40	63.06%	51.58%	28.68%	34.69%	8.26%	13.73%
Computer	41.09	55.56	64.88%	54.16%	27.62%	33.57%	7.50%	12.27%
Imaging	47.81	64.70	60.51%	49.85%	30.26%	35.59%	9.23%	14.56%
Mobile phone	49.58	67.61	56.58%	45.06%	32.39%	37.41%	11.02%	17.53%
Phone	54.77	74.30	54.29%	43.19%	33.56%	38.07%	12.15%	18.74%
TV	42.21	57.40	65.76%	54.69%	27.11%	33.35%	7.13%	11.97%
TV box	49.34	66.66	57.96%	47.28%	31.66%	36.61%	10.38%	16.12%
Whole sample	49.33		58.99%		30.86%		10.15%	

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is a very complex decision. Still, our results can inform that decision with new empirically grounded insights into experience durations by product category.

Last, we profile the newness of returns based on our estimation results. Suppose a retailer knows from past experience that returns made within the first 7 days can be restocked as open-box items, which is the most cost-effective salvaging option. In addition, the OEM may provide a full refund to the retailer for all products returned within 30 days, which is the second most cost-effective salvaging option for the retailer. For returns older than 30 days, the retailer has to sell them to a third-party refurbisher. In order to allocate the necessary resources for each of these salvage options, the retailer needs an estimate of the proportion of returns that fall within each time interval. We perform the estimation of these proportions for each category-holiday combination and the whole sample (results given in Table 8). For this retailer, the proportion of "new" returns (lag < 7 days) is between 50% and 60% during the nonholiday time, while it is between 45% and 55% during the holiday period, while the proportion of "old" returns (lag > 30 days) makes up around 10% of all returns during nonholiday time and around 15% during holidays.

CONCLUSION

It has been estimated that defective products only make up 5% of all consumer electronic returns (Douthit et al., 2011). The remaining 95%, mainly due to product mismatches and buyer remorse, are controllable to a certain extent by retailers. Our study contributes to the management of these controllable returns.

Theoretical Implications

Several of the key findings in this article have direct implications for analytical work on consumer returns. For example, the correlation between product maturity and the probability of a return estimated by our model is informative for multiperiod product return models. In a two-period analysis, Ketzenberg and Zuidwijk (2009) assume that consumers are equally capable of evaluating a product in the first period versus in the second period. Our results, which show a small within-product effect of maturity (see model R3 in online Appendix B), suggest that this is a reasonable assumption. We also add to the product variety literature by showing the return reduction benefit of adding more variety in a product assortment. This benefit can be included with the typical inventory cost and consumer demand tradeoff involved in product assortment decisions (van Ryzin & Mahajan, 1999; Tan, Netessine, & Hitt, 2015).

Prior empirical studies show a positive future demand effect from customers who make more returns (e.g., Petersen & Kumar, 2009; Griffis et al., 2012). We add to this discussion by showing that customers with more past return instances not only have a higher return probability, but also tend to keep products longer before returning them—both factors increasing a retailer's cost to serve them. Therefore, retailers should take a more holistic view, weighing both future demand and future returns to determine the overall profit impact of these return-prone consumers.

Finally, we also replicate some return-related effects reported in the literature. For example, confirming Petersen and Kumar (2009), we also find that products

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purchased during holidays and on sale are more likely to be returned. Similar to Hess and Mayhew (1997), our analysis also shows that the price of a product does not have an impact on the timing of return.

Managerial Implications

From a practical perspective, our findings in this article are relevant for retailers who seek to employ buyer assistance programs to reduce their volumes of returns by helping consumers resolve their fit uncertainty. The hope in offering these programs is that the cost of providing them will be offset by a drop in the return rates (as well as potential increases in sales and customer satisfaction). Given limited staff resources and floor space, it is not possible for retailers to provide all of these buyer assistance services for their entire product assortment. Our findings suggest that these buyer assistance investments provide larger benefits when applied to newer products as well as products that have fewer competing alternatives. Even if a more targeted buyer assistance program only results in a 1% decrease in the return rate, the National Retail Federation (2018) estimates that, when accounting for the costs incurred by various parties along the reverse supply chain, this decrease can result in $31.9 billion in savings for the U.S. economy. Therefore, we believe our insights for better targeting of buyer assistance programs have strong practical significance.

The National Retail Federation (2018) reports that a quarter of the returns in 2017 were made during the holiday period. Despite this high volume, our analysis on the time dimension of returns suggests that, due to longer experience duration, the return time window should be longer for holiday purchases. This finding echoes practices from leading retailers: in the 2017 holiday season, Amazon extended its normal 30-day return window for purchases made in November–January 31, 2018, while Apple made a similar extension to January 8, 2018 on its typical 14-day window (Pearson, 2017). In addition, although we observe some degree of return time window variation across retailers, such as 45 days for Gap (apparel) and 15 days for BestBuy (electronics), most retailers still use a uniform window across all of their products. Our results on the categorical heterogeneity of experience duration suggest that retailers might consider a more fine grained customization of the return time window across product categories. This call for action is shared by return management practitioners (Gilbreath, 2017).

Future Research

Although the literature on consumer returns is growing rapidly, empirical work in this area remains scarce. One promising direction for future empirical studies is to measure the effectiveness of buyer assistance programs directly. While our results on the drivers of return probability can guide where to target these efforts, field experiments can provide more direct measurements of the impact of targeted assistance. In practice, buyer assistance takes various forms. For example, sales personnel might spend extra time with consumers and listen to their needs thoroughly before recommending a product (Ferguson et al., 2006). In addition, more samples may be displayed on the floor and the full array of similar products may be arranged more effectively on the shelf (Ofek et al., 2011). Each of these treatments

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is likely to be more effective for certain products. Studying the heterogeneity of their effectiveness may provide valuable implications for retailers. Such analysis can also be extended to online retailers to help study the return reduction effectiveness of recent virtual fitting technologies (Gallino & Moreno, 2018). Our insights in this article can serve as the foundation for these and other empirical analyses of the drivers of consumer returns.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Supporting Information

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Guangzhi Shang is an assistant professor and the Dean's Emerging Scholar in the Department of Analytics, Information Systems, and Supply Chain at Florida State University. His research has been published in *Production and Operations Management* and *Journal of Operations Management*, among others. His research on the value of the return policy is the finalist for the 2018 *Journal of Operations Management* best article award. He is also recognized by the 2018 best reviewer award of the *Journal of Operations Management* and the nomination of

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the 2017 best reviewer award of the *Production and Operations Management Journal*. Guangzhi Shang's current research has two primary themes: consumer returns management and service labor issues. He investigates the former from a variety of angles, including how a retailer should set its optimal return policy, how an OEM or a retailer could better forecast the quantity of returns, and how a retailer could assess the value of its return policy. For the latter, he focuses on the context of live-chat contact centers. Research questions include the impact of customer's waiting experience on the progress of a chat session, agent's ability to learn from their past experiences, and the customer–agent matching problem. He enjoys doing practice-driven research. He is a frequent invited speaker at leading industry conferences such as the annual Consumer Returns conference. He also coproduces a column in the Reverse Logistics Magazine named "View from Academia," aimed at disseminating fresh-off-the-press academic knowledge among industry professionals dealing with consumer returns.

Mark Ferguson is a Distinguished Business Foundation Fellow and professor of Management Science in the Darla Moore School of Business at the University of South Carolina. His research interests involve many areas of supply chain management including supply chain design for sustainable operations, contracts that improve overall supply chain efficiency, pricing and revenue management, and the operations/marketing interface. His 2012 article on the environmental impact of product leasing won the best operations management article award for articles published between 2012 and 2014 in *Management Science*. Another two of his articles have won best article awards from the Production and Operations Management Society (POMS), and three of his research projects have been funded by the National Science Foundation. He is the coauthor of the books *Segmentation, Revenue Management and Pricing Analytics and Pricing Segmentation and Analytics* and coeditor of the book *Closed Loop Supply Chains: New Developments to Improve the Sustainability of Business Practices*. He has served as the president of the INFORMS Manufacturing and Services Operations Management Society, the president of the POMS College of Supply Chain Management and the INFORMS Revenue Management and Pricing Section. Prior to joining the Moore School in 2011, he was the Steven Denning Professor of Technology and Management at the College of Management at the Georgia Institute of Technology and spent five years as a manufacturing engineer and inventory manager with IBM.

Michael Galbreth is a Professor and Head of the Department of Business Analytics and Statistics at the Haslam College of Business, University of Tennessee, where he also holds the Pilot Corporation Chair of Excellence. He received his PhD from Vanderbilt University. Prior to joining the Haslam College, he was a Professor of Management Science at the University of South Carolina, where he taught a wide variety of MBA and undergraduate courses and was a seven-time winner of the IMBA outstanding professor award. His research activities are currently focused on retail operations, consumer returns, sustainable operations, and the interface between marketing and operations. In 2010, he was named a Fulbright Scholar and served as the Fulbright Enders Visiting Research Chair at McGill University. In 2014, he received the Paul Kleindorfer Award in Sustainability

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from the Production & Operations Management Society. He is a Department Editor for *Journal of Operations Management*, a Senior Editor for *Production & Operations Management*, and an Associate Editor for *Decision Sciences*. In addition to *Decision Sciences*, his research has also appeared in *Management Science*, *Marketing Science*, *MIS Quarterly*, *Journal of Operations Management*, *Production & Operations Management*, *Interfaces*, and others. He has consulted with a wide variety of companies, with a focus on building analytical decision models and developing analytics training materials and programs. Prior to returning to academia, he was a manager in the Atlanta office of KPMG Consulting.

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